A L I A L - K E L A B I



Personal Data

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About

AI and mechatronics engineer focused on building practical systems across AI workflows, automation, data pipelines, desktop software, embedded systems, and robotics. I develop structured, usable solutions that combine technical depth with clear architecture, fast debugging, and real-world relevance. I quickly familiarize myself with new tools and use AI-powered tools strategically to accelerate development, research, and debugging, without losing sight of quality. Diligence, attention to detail, and reliability are deeply ingrained in my projects, and I am accustomed to reliably balancing my studies with practical work.

Core skills

- AI & ML: LLM applications, RAG, explainable AI, human-in-the-loop systems, computer vision, anomaly detection, classification workflows.
- Software Engineering: Python, C++, FastAPI, Flask, Qt, PyQt6, Streamlit, React, Vite, SQL, REST APIs, modular architecture.
- Data & Analytics: BigQuery, SQL Server, PySpark, data pipelines, KPI dashboards, ingestion workflows, validation, logging, reporting.
- Automation: UiPath, web scraping, workflow automation, reusable technical processes, internal productivity tools.
- Robotics & Vision: ROS, Gazebo, MoveIt, OpenCV, MediaPipe, perception pipelines, gesture-based systems, robot workflow integration.
- Embedded & Systems: Raspberry Pi, LoRa, sensor integration, edge systems, GUI monitoring tools, hardware-software integration, communication workflows
- Languages: German (fluent, C1), English (business fluent, C2), Arabic (native)

Work Experience

Student Assistant – UiPath & C# Automation of Computer Tasks (June 2025 – November 2025)
Deggendorf University of Applied Sciences, Cham, Germany

- Development of a UiPath automation course featuring practical use cases.
- Implementation of web scraping solutions for automated data pipelines.
- Supporting process optimization through automation and structured technical implementation.

Project Collaboration – Deep Learning - Machine Learning - Computer Vision (March 2024 – July 2024)
ICON, Malaysia

- Collaboration on an ML project (action recognition) and DL (YOLO) to detect defined violations in public spaces, with relevant information—including license plate and route data—transferred to database structures for follow-up.
- Collaboration on a second use case (OCR) for the detection of locomotives, extraction of visible information and size-related features for tracking purposes.
- Support for Python-based training and data preparation processes, as well as the use of labeling tools such as CVAT and LabelImag.

Working Student – ML & Sensor Data Analysis (Nov 2019 – Feb 2023)
IIUM & Malaysian Nuclear Agency, Malaysia

- Development and adaptation of 3D designs for PEC probes in the context of non-destructive material testing.
- Conducting pulsed eddy current tests on ferromagnetic test plates using an electrical measurement chain and LabVIEW interface.
- Analysis of recorded signals for the detection and evaluation of corrosion under insulation.
- Collaboration on research projects related to signal processing and material inspection.

Projects

AML Alert Triage Copilot (May 2026 – June 2026)

- Built an AI-assisted AML investigator copilot that ingests transaction-monitoring alerts as JSON and generates a structured triage summary with suspected typology, risk bucket, risk indicators, false-positive signals, next action, explanation, and confidence.
- Combined an LLM backend with a Python rule overlay to enforce safer decisions and improve explainability, reflecting a rules-plus-explainable-AI workflow similar to modern AML platforms.
- Designed the system as explicitly human-in-the-loop: the model supports triage and reasoning, while the final investigator decision remains with the user.
- Implemented a demo-ready workflow with FastAPI, Pydantic schemas, Groq-hosted Llama 3, and a Streamlit interface for review, editing, and simulated case saving.

CampaignPulse – Analytics Dashboard for Purchase Tracking and KPI Reporting (April 2026 – May 2026)

- Development of a product-oriented analytics dashboard using React, Vite, FastAPI, Recharts, and BigQuery to analyze purchase, funnel, and KPI data.
- Implementation of a clearly separated architecture comprising frontend, API routing, service layer, and cloud-based data querying with mock-capable fallback for local development.
- Creation of a professional-looking interface for non-technical stakeholders with a focus on dashboard behavior rather than pure data output.

Industrial Telemetry Ingestion Desktop App (April 2026 – May 2026)

- Development of a C++/Qt desktop application for ingesting industrial telemetry data from CSV files into Microsoft SQL Server via ODBC.
- Implementation of GUI, validation logic, reject logging, reset/refresh workflows, and robust data ingestion for technical operations.
- Focus on fault-tolerant data processing, usability, and transparent system states in desktop environments.

Gesture-Based Access Control System (Mar.2026 - April.2026)

- Development of a gesture-based access system featuring a local GUI, MediaPipe hand landmarks, a GRU classifier, a Flask API, and HTTP-based inference.
- Implementation of sequence logic, landmark processing, GUI/API separation, and a reproducible setup for real-time use.
- Focus on the technical integration of computer vision, ML inference, and access control logic.

Local Research Assistant (Feb.2026 - Mar.2026)

- Development of a local PyQt6 desktop application for document analysis using Qwen2.5-RAG, with PDF/TXT upload and fully offline inference capabilities.
- Responsible for system design, GUI, local LLM integration, encrypted history data, and production-ready desktop architecture.
- Result: 60–65 tokens/s CPU inference, local multi-PDF analysis, no cloud dependency after model download.

Spark Mastery – Production Projects (Feb.2026 - Mar.2026)

- Development of Dockerized PySpark pipelines for anomaly detection across multiple domains, including IoT, e-commerce, server logs, and social media.
- Processing of over 7.2 million data records using Z-score-based detection and production-ready execution in Spark 3.5.1.
- Result: Over 299,000 detected anomalies across multiple data domains, including a reproducible pipeline structure.

UR5 Vision-Guided Pick-and-Place (Oct.2025 – Jan.2026)

- Technical revision, structuring, and ensuring the reproducibility of a ROS-based UR5 pick-and-place system using Gazebo, MoveIt, and OpenCV
- Contribution to system design, architecture, setup structure, documentation, debugging, and an executable development environment in a Docker container
- Implementation of a vision-guided process for detecting and sorting red objects using perception, motion planning, and control
- Result: 98.2% success rate over 100 cycles, approx. 523 ms planning time, 1.43 s cycle time, vision accuracy 99.3%

Automated Diabetic Retinopathy Severity Grading (Oct.2025 - 01.2026)

- Development of an end-to-end deep learning pipeline for automated severity assessment of diabetic retinopathy using ResNet-50 and DenseNet-121
- Work on preprocessing, model training, evaluation, and comparison of classification versus ordinal regression
- Result: DenseNet-121 with an AIC of approx. 0.78–0.79 and improved detection of referable DR cases

Education

M.Eng. in Artificial Intelligence for Smart Sensors and Actuators (March 2025 – Present)

Deggendorf Institute of Technology (THD), Cham, Germany

- Focus areas: Machine Learning, Deep Learning , Computer Vision, System Design, Edge Devices, Advanced Programming, Big Data
- Grade point average (German system): 2.4

B.Eng. in Mechatronics, with honors (Jan. 2019 – Feb. 2023)

International Islamic University of Malaysia (IIUM), Gombak, Malaysia

- Major areas of focus: Automation, Mathematics, Robotics, Software Engineering, Machine Learning, Deep Learning , Computer Vision, Intelligent Control
- GPA (German grading scale): 2.0
- Bachelor’s Thesis / Research Context: Contribution to research in the field of non-destructive testing and corrosion detection using pulsed eddy current, including signal processing, sensor testing, and 3D sample development.

Availability

- Available immediately as a working student during the semester and also during the semester break.
- Enrolled at a German university.